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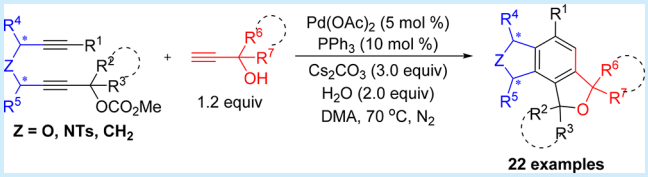
Construction of Benzopolycycles via Pd-Catalyzed Intermolecular Cyclization of 2,7-Alkadiynylic Carbonates with Terminal Propargyl Tertiary Alcohols

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Supporting Information

ABSTRACT: A palladium-catalyzed highly regioselective and chemoselective intermolecular cyclization of 2,7-alkadiynylic carbonates with terminal propargyl tertiary alcohols to construct benzopolycycles containing furan and pyrrole moieties has been developed with a very broad scope. Polycycles containing spirane structures or dispirane structures could be also smoothly synthesized in moderate to good yields. The reaction enjoys excellent regioselectivity.



Benzopolycycles containing furan and pyrrole moieties are important skeletons in potential pharmaceutical molecules (Figure 1).[1] For example, concentricolide is a compound with

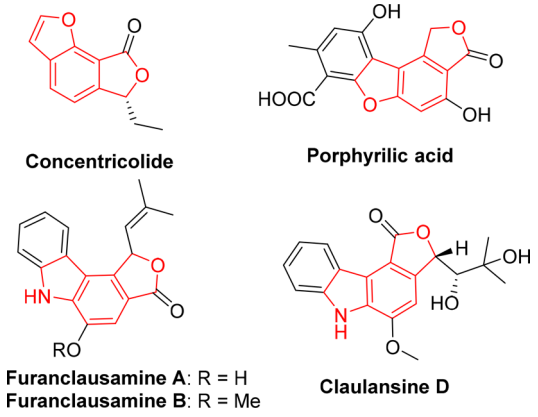
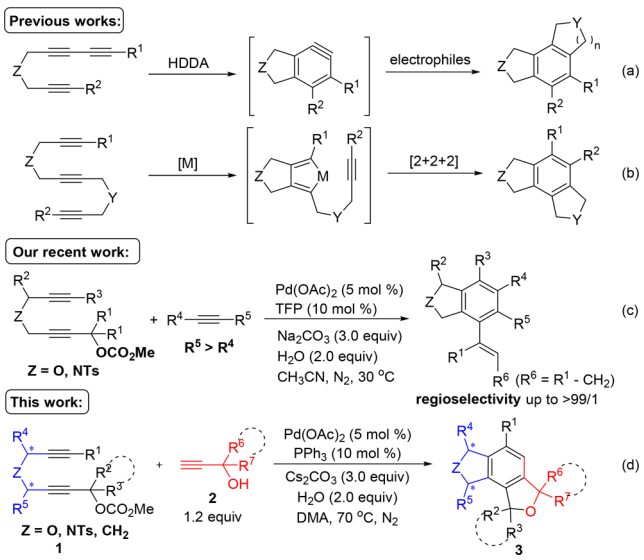


Figure 1. Some typical pharmaceutical benzopolycycles containing furan and pyrrole moieties.

in vitro activity against HIV-1[1c] and furanclausamine **A** as well as **B** have anti-bacterial activity.[1d,e] These types of skeletons were commonly constructed from linear triynes: (1) the intramolecular Hexadehydro-Diels–Alder (HDDA) reaction of 1,3,8-triynes forms aryne intermediates, which subsequently react with nucleophiles to produce tricyclic molecules (Scheme 1a);[2,3] (2) the transition metal-catalyzed intramolecular [2 + 2 + 2] cyclization of 1,6,11-triynes (Scheme 1b).[4,5] In addition, tandem reactions consisting of oxidative addition, consecutive carbopalladation, and eventual Heck process of alkenyl halides with diynes were also efficient routes to benzopolycycles.[6] Recently, our group developed an efficient process of benzofuran construction via a Pd-catalyzed intermolecular annulation of 2,7-alkadiynylic carbonates with alkynes (Scheme 1c).[7] Interestingly, when we applied propargyl tertiary alcohols to this reaction, the chemoselectivity was unexpectedly different: tricyclic skeletons could be easily constructed via the intramolecular nucleophilic attack of oxygen anion instead of the β-H elimination. In this paper, we wish to report our such recent observations (Scheme 1d).

Initially, we conducted this reaction of 2,7-alkadiynylic carbonate **1a** and 2-methylbut-3-yn-2-ol **2a** (1.2 equiv) under the standard conditions reported in our recent work.[7] After

Scheme 1. Routes for the Construction of Benzopolycycles



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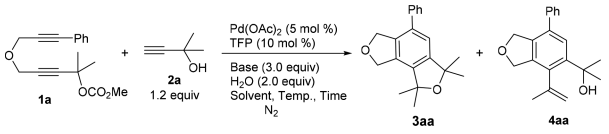
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~45 h, tricyclic benzodifuran derivative **3aa** was observed unexpectedly in 48% yield, together with the expected benzofuran derivative **4aa**[7] in 23% yield (entry 1 in Table 1). Considering the importance of benzodifuran derivatives,[1] we set out to investigate the influence of the critical reaction parameters (Table 1). First, the reactions failed to give better results when NaOH, K_2CO_3, and K_3PO_4 were used in place of Na_2CO_3 (entries 2–4 in Table 1). However, the reaction with Cs_2CO_3 as the base afforded **3aa** in 54% yield (entry 5 in Table 1). Raising the temperature resulted in a slightly better yield of **3aa** (entries 6 and 7 in Table 1). Notably, the reaction in DMA gave a quite decent selectivity of **3aa**/**4aa**, although the yield was slightly lower than that in CH_3CN after the screening of a series of solvents such as DMF, DMA, NMP, and DMSO (entries 8–11 in Table 1).

Table 1. Effects of Base, Solvent, and Temperature[a]



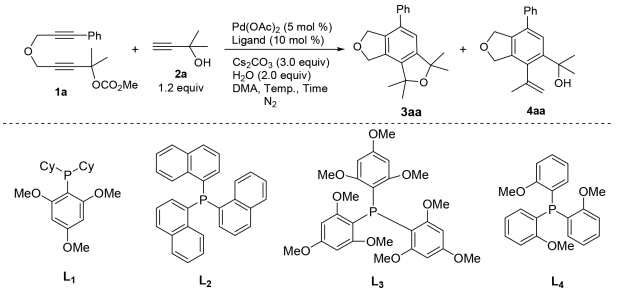
entry	base	solvent	temperature (°C)	time (h)	yield of **3aa**/**4aa**[b] (%)
1[c]	Na_2CO_3	CH_3CN	30	~45	48/23
2	NaOH	CH_3CN	30	25.2	13/12
3	K_2CO_3	CH_3CN	30	25.9	42/26
4	K_3PO_4	CH_3CN	30	24.7	46/29
5	Cs_2CO_3	CH_3CN	30	24.5	54/21
6	Cs_2CO_3	CH_3CN	50	21.8	58/12
7	Cs_2CO_3	CH_3CN	70	21.4	58/18
8	Cs_2CO_3	DMF	50	22	44/3
9	**Cs_2CO_3**	**DMA**	**50**	**22**	**55/3**
10	Cs_2CO_3	NMP	50	22	21/1
11	Cs_2CO_3	DMSO	50	22	12/0

[a]Reaction conditions: **1a** (0.2 mmol), **2a** (1.2 equiv), $Pd(OAc)_2$ (5 mol %), TFP (Tri(2-furyl)phosphine) (10 mol %), base (3.0 equiv), H_2O (2.0 equiv) in solvent (2 mL) under N_2 atmosphere unless otherwise noted. [b]Determined by ^1H NMR analysis of the crude product using mesitylene as an internal standard. [c]The reaction was conducted on 1.0 mmol scale of **1a**.

Further screening of monophosphine ligands for the reaction in DMA showed that the reactions with **L_1**•HBF_4,[8] **L_2**, **L_3**, **L_4** or $P(Cy)_3$ all delivered worse results except the reaction with PPh_3 as the ligand, which produced **3aa** in 59% yield exclusively (entries 1–6, Table 2). After a test of the temperature, it was exciting to find that the yield of **3aa** could be increased to 77% at 70 °C (entries 8, Table 2). Higher temperature seemed to be invalid to improve the yield (entries 9 and 10, Table 2). Above all, the optimal conditions were defined as follows: $Pd(OAc)_2$ (5 mol %), PPh_3 (10 mol %), Cs_2CO_3 (3.0 equiv) and H_2O (2.0 equiv) in DMA at 70 °C.

The reactivity of various oxygen-tethered 2,7-alkadiynylic carbonates was examined with differently substituted terminal propargyl tertiary alcohols on a 1 mmol scale (Scheme 2). In addition to the methyl group, R^5 could represent all of the other alkyl groups, such as isobutyl, *n*-pentyl, and *n*-nonyl. The reactions afforded benzodifurans products **3aa**–**3ad** in 67%–81% yields. R^4 and R^5 groups could also be cyclic groups with different sizes, ranging from 4-membered ring to 12-membered ring. The desired polycycles containing a spirane structure

Table 2. Effects of Ligand and Temperature in DMA[a]



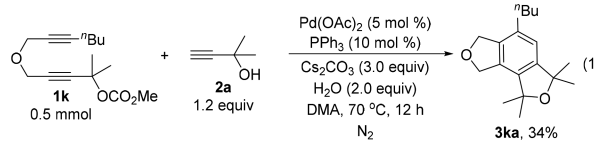
entry	ligand	temperature (°C)	time (h)	yield of **3aa**/**4aa**[b] (%)	recovery of **1a**[b] (%)
1	$P(Cy)_3$	50	22.1	21/0	55
2	**L_1**·HBF_4	50	22	8/0	0
3	**L_2**	50	22	3/0	33
4	**L_3**	50	22	0/0	0
5	**L_4**	50	22	15/0	12
6	PPh_3	50	22	59/0	1
7	PPh_3	60	22	63/0	0
8	**PPh_3**	**70**	**22**	**78/0**	**0**
9	PPh_3	80	19	76/0	0
10	PPh_3	90	18.5	77/0	0

[a]Reaction conditions: **1a** (0.2 mmol), **2a** (1.2 equiv), $Pd(OAc)_2$ (5 mol %), ligand (10 mol %), Cs_2CO_3 (3.0 equiv), H_2O (2.0 equiv) in DMA (2 mL) under N_2 atmosphere, unless otherwise noted. [b]Determined by ^1H NMR analysis of the crude product using mesitylene as the internal standard.

were produced smoothly in 69%–77% yields (**3ae**–**3ch**). The reaction of **1a** and **2e** could be easily conducted on a gram scale, delivering 70% yield of **3ae** and 8% yield of **4ae**, respectively. The 2,7-alkadiynylic carbonates with the aryl group bearing either electron-donating or electron-withdrawing groups at the meta position or the para position could all be applied satisfactorily, affording polycyclic products **3bg**–**3eh** in good yields. The structure of **3eh** was confirmed by its single-crystal X-ray diffraction analysis (Figure 2). In addition, this method could be extended to 8-(3′-thienyl)-substituted 2,7-alkadiynylic carbonate (**3fh**).

The scope of R^2 and R^3 was next examined with propargyl tertiary alcohol (**2a** or **2h**). The reactions proceeded smoothly, affording the corresponding polycycles **3ga**–**3ja** in moderate to good yields, whether the R^2 and R^3 groups were identical or not. Note that the pentacyclic product **3jh** containing a dispirocyclic structure could be obtained in 52% yield together with 6% yield of the *β*-H elimination byproduct **4jh**.

Furthermore, the 8-^{n}Bu-substituted 2,7-alkadiynylic carbonate **1k** was also compatible in this reaction, although the yield was only 34% (see eq 1):

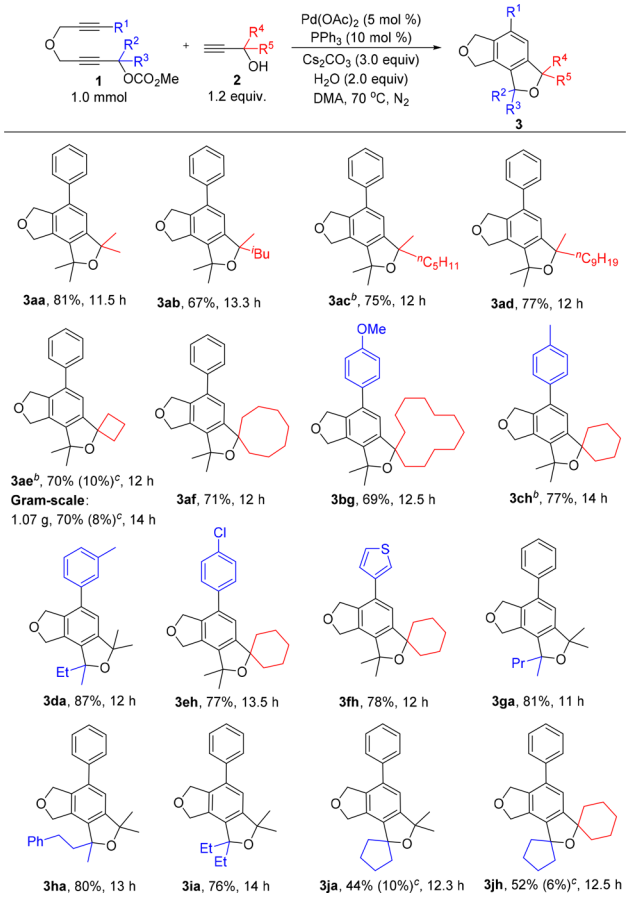


Encouraged by the results of oxygen-tethered 2,7-alkadiynylic carbonates, we attempted to introduce carbon- or nitrogen-tethered 2,7-alkadiynylic carbonates for this reaction. The corresponding products, indeno[4,5-*c*]furan derivative (**3la**) and furo[3,4-*e*]isoindole derivative (**3 mi**), were isolated in moderate yields (Scheme 3).

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Scheme 2. Scope of 2,7-Alkadiynylic Carbonates (1) and Terminal Propargyl Tertiary Alcohols (2)[a]



[a]Reaction conditions: **1** (1.0 mmol), **2** (1.2 equiv), $Pd(OAc)_2$ (5 mol %), PPh_3 (10 mol %), Cs_2CO_3 (3.0 equiv), and H_2O (2.0 equiv) in DMA (10 mL) at 70 °C. [b]H_2O (2.1 equiv). [c]Isolated yield of the β-H elimination **4**-type byproducts given in parentheses.

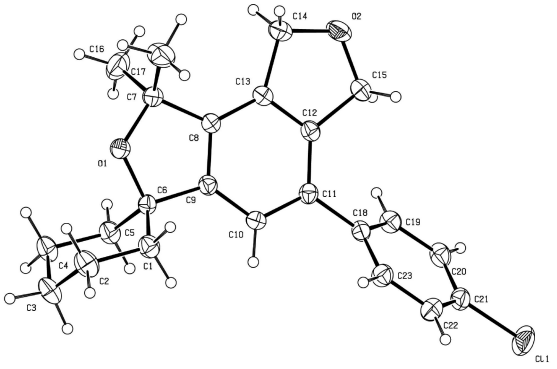
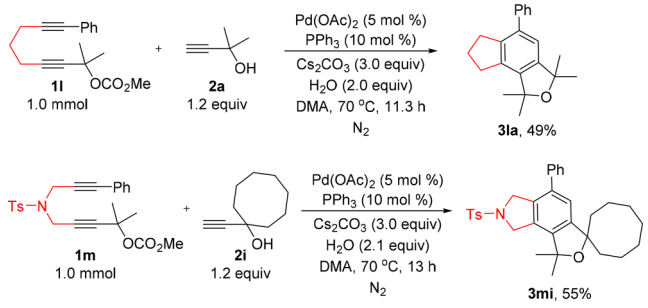


Figure 2. ORTEP representation of **3eh**.

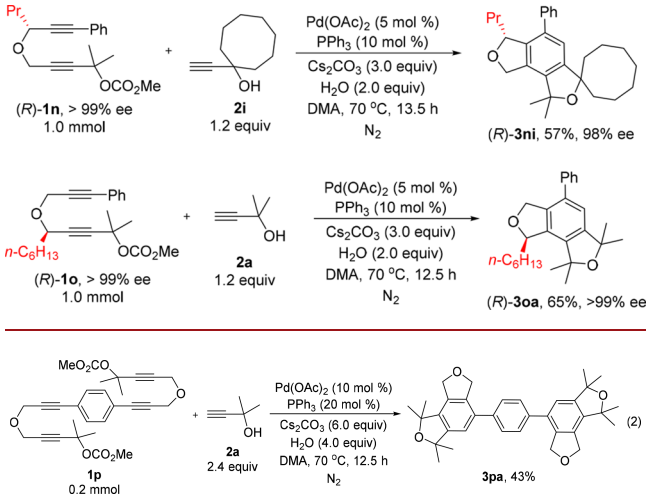
Furthermore, when the optically active 2,7-alkadiynylic carbonates (*R*)-**1n** and (*R*)-**1o** were loaded under standard conditions, (*R*)-**3ni** and (*R*)-**3oa** were obtained in 57% and 65% yields, respectively, with no loss of enantiomeric purity (Scheme 4).

Interestingly, a symmetric polycycle **3pa** could be formed in 43% yield with six rings constructing in just one synthetic operation from the substrate **1p**, which has two symmetric yne-propargylic carbonate units (see eq 2).

Scheme 3. Reaction of Carbon- or Nitrogen-Tethered 2,7-Alkadiynylic Carbonates (1) and Propargyl Tertiary Alcohols (2)

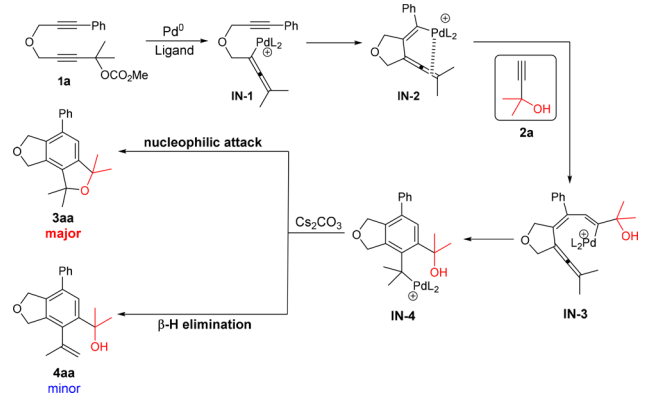


Scheme 4. Synthesis of Optically Active Products (*R*)-3ni and (*R*)-3oa



Based on the experimental results and our previous works,[7] a possible mechanism is shown in Scheme 5 by taking the reaction of **1a** and **2a** as the typical example: oxidative addition of **1a** with Pd(0) coordinated with PPh_3 would give the allenylpalladium intermediate **IN-1**,[9] which underwent intramolecular insertion to deliver the alkenylpalladium species **IN-2**.[10] Because of the strong steric effect of the propargylic carbon atom, the highly regioselective intermolecular insertion of **IN-2** and propargyl tertiary alcohol **2a** would generate **IN-3**,[7] which underwent intramolecular insertion of the 4-position C═C bond of the allene moiety to give a benzylpalladium

Scheme 5. Proposed Mechanism



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intermediate **IN-4**. In the presence of Cs_2CO_3, the key intermediate **IN-4** could undergo the intramolecular nucleophilic attack or β-H elimination, forming tricyclic product **3aa** or the phthalan derivative **4aa** and regenerating the catalytically active Pd(0). The high chemoselectivity of **3aa**/**4aa** should also depend on the steric hindrance of the tertiary alcohol, which made the hydroxy group close enough to coordination to the palladium to avoid β-elimination.

In summary, we have established a highly chemoselective annulation of 2,7-alkadiynylic carbonates with terminal propargyl tertiary alcohols to generate benzodifuran, furo[3,4-*e*]isoindole, and indeno[4,5-*c*]furan derivatives. Three new C−C bonds and a C−O bond are formed in just one synthetic operation. Polycycles containing spirane structures with different size could be easily prepared from readily available cyclic alkyl substituted propargyl alcohols. The central chirality in the starting materials could be retained in the products without erosion of enantiomeric excess. Further studies on synthetic applications and biological activity of these products are being pursued in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.9b00746.

Detailed experimental procedures; characterization data of all of the new compounds; copies of HPLC chromatographies; ^{1}H and ^{13}NMR spectra of the products (PDF)

Accession Codes

CCDC 1870594 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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